



WANDERLUST

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Slot: L4+L5

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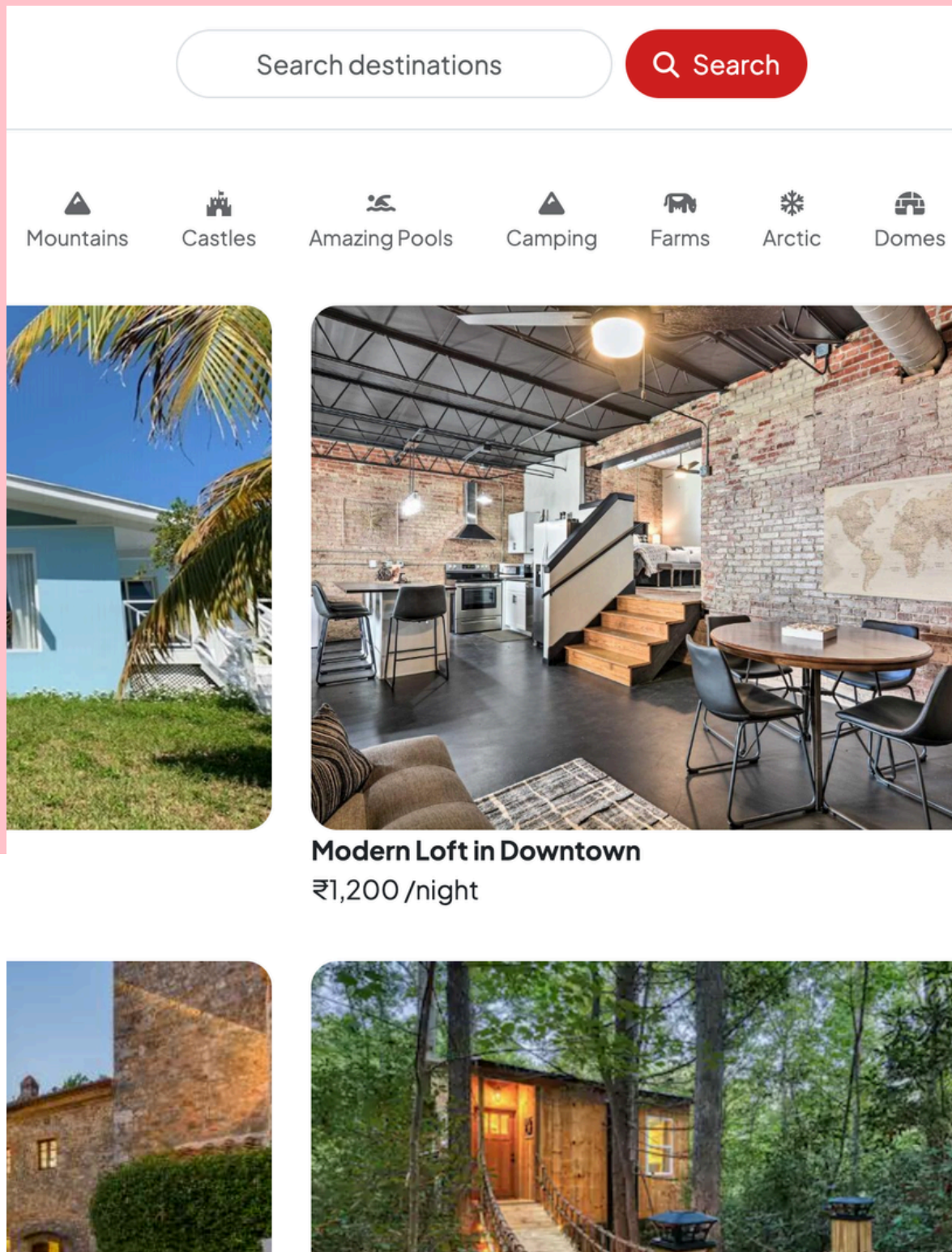
Problem Statement

The current state of travel planning is fragmented and inefficient.

Travelers often rely on multiple platforms to find hotels, plan itineraries, and discover nearby attractions. This process can be time-consuming and frustrating, leading to suboptimal travel experiences.

Our project aims to address these challenges by providing a comprehensive, user-friendly platform for travel planning. We will integrate hotel booking functionalities, real-time maps, and curated lists of nearby tourist attractions to streamline the entire travel planning process. By offering a one-stop solution, we aim to simplify travel planning and enhance the overall traveler experience.

Abstract



- **‘Wanderlust’ is a cutting-edge software platform developed to innovate the experience of travel accommodation by making available an all-in-one, easy-to-use platform for discovering and booking one-of-a-kind stays anywhere in the world. With the very successful model of Airbnb as a reference, Wanderlust users can discover and reserve rich variations of accommodations—from a cozy apartment to a luxurious villa or unique and adventurous stays in exotic locations.**
- **High levels of integration in terms of travel convenience features set Wanderlust apart. It provides information to users on transportation and available amenities, like real-time fares and distances to key destinations that people normally look for from an area or accommodation. These include the nearest bus and railway stations, airports, key restaurants, and the like. This helps individuals and groups plan their journeys in terms of travel routes easily and saves time in logistics.**
- **Wanderlust ensures that users can select options suiting their priorities and needs by providing an overview of travel costs and distances from important and popular destinations. Besides, planning travel becomes easy and fun due to the intuitive and user-friendly interface of the platform and the search functionality across it.**
- **Wanderlust serves all kinds of needs that modern travelers may have by providing a reliable and efficient tool for discovering new destinations or returning to favorite places in the simplest possible way. Since it is committed to innovation and customer satisfaction, Wanderlust has set a new standard for the travel and accommodation industry by giving users the power to create unforgettable and trouble-free traveling memories.**

Key Features

Map API Integration

Wanderlust leverages a robust Map API to provide interactive maps, location-based search, and real-time distance calculations, enhancing user experience and decision-making.

Dynamic Pricing and Recommendations

The platform utilizes advanced algorithms to offer dynamic pricing based on demand, seasonality, and user preferences, providing personalized accommodation recommendations.

Immersive Virtual Tours

To enhance the booking experience, Wanderlust offers high-quality virtual tours of accommodations, allowing users to explore properties in detail before making a decision.

AI-Powered Chatbots

Implement AI-powered chatbots to provide instant customer support, answer queries, and assist with bookings.

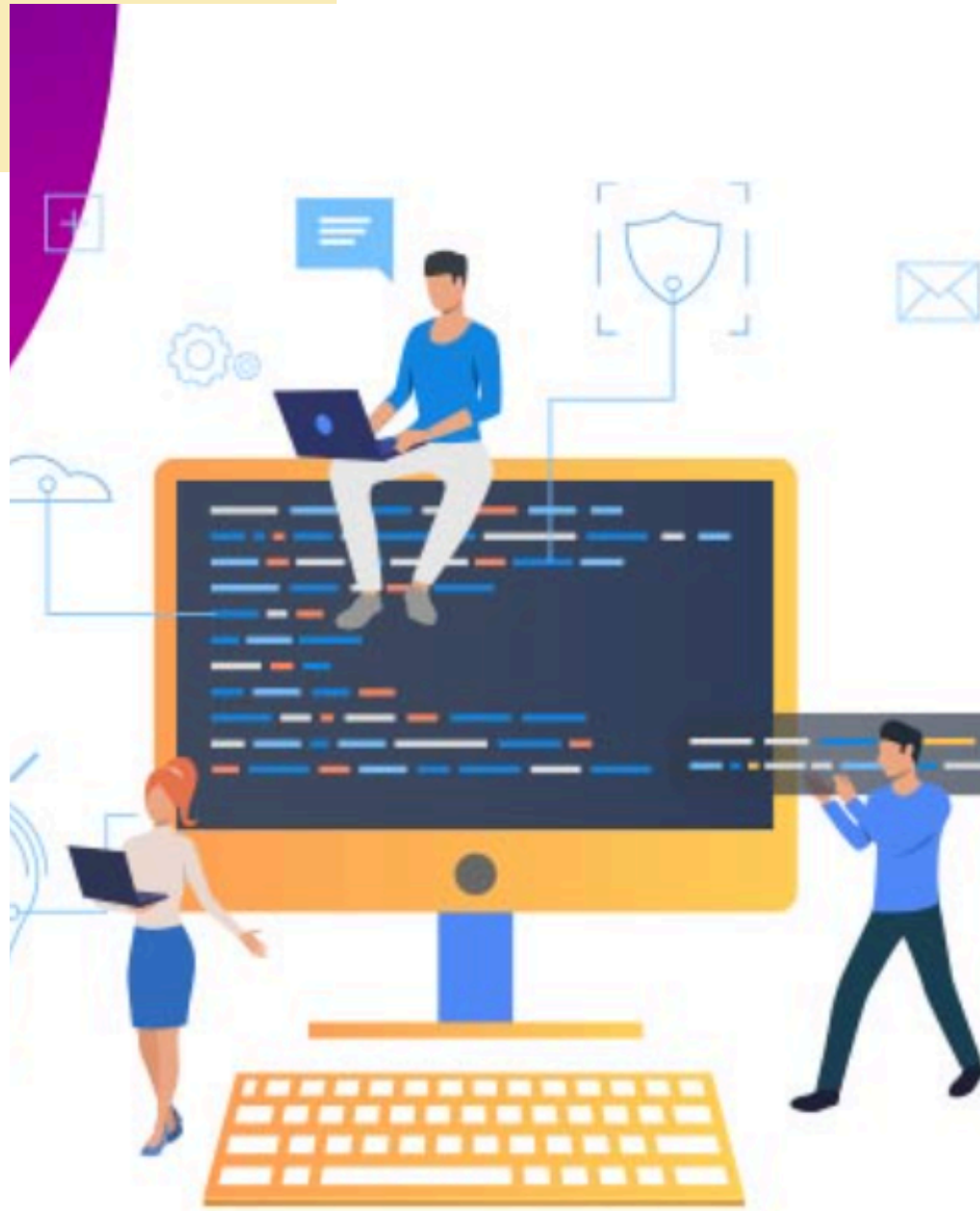
Hardware Requirements

User-Side Requirements:

- 1) Any modern computer with a functioning web browser or smartphone or tablet with modern web browser.
- 2) User need a stable and reasonably fast internet connection to access the web application efficiently, especially for viewing images and maps.

Server-Side Requirements:

- 1) Minimum Intel i5 or equivalent; Intel i7 or higher recommended for faster performance.
- 2) Minimum of 8 GB, with 16 GB or more recommended for handling more traffic and ensuring smooth operation.
- 3) SSD with at least 250 GB of storage, scalable based on the number of users and amount of data (especially if media like images are stored on the server).
- 4) High-speed internet connection with sufficient bandwidth to handle user traffic and data transfer, especially for image loading and API requests.



Software Requirements

Node.js: Utilized for the server-side, providing a robust runtime environment to handle backend operations efficiently.

MongoDB: A NoSQL database, perfect for storing and managing the application's data dynamically.

npm Library: Essential for managing and installing various packages and dependencies, ensuring the project has the necessary tools and libraries.

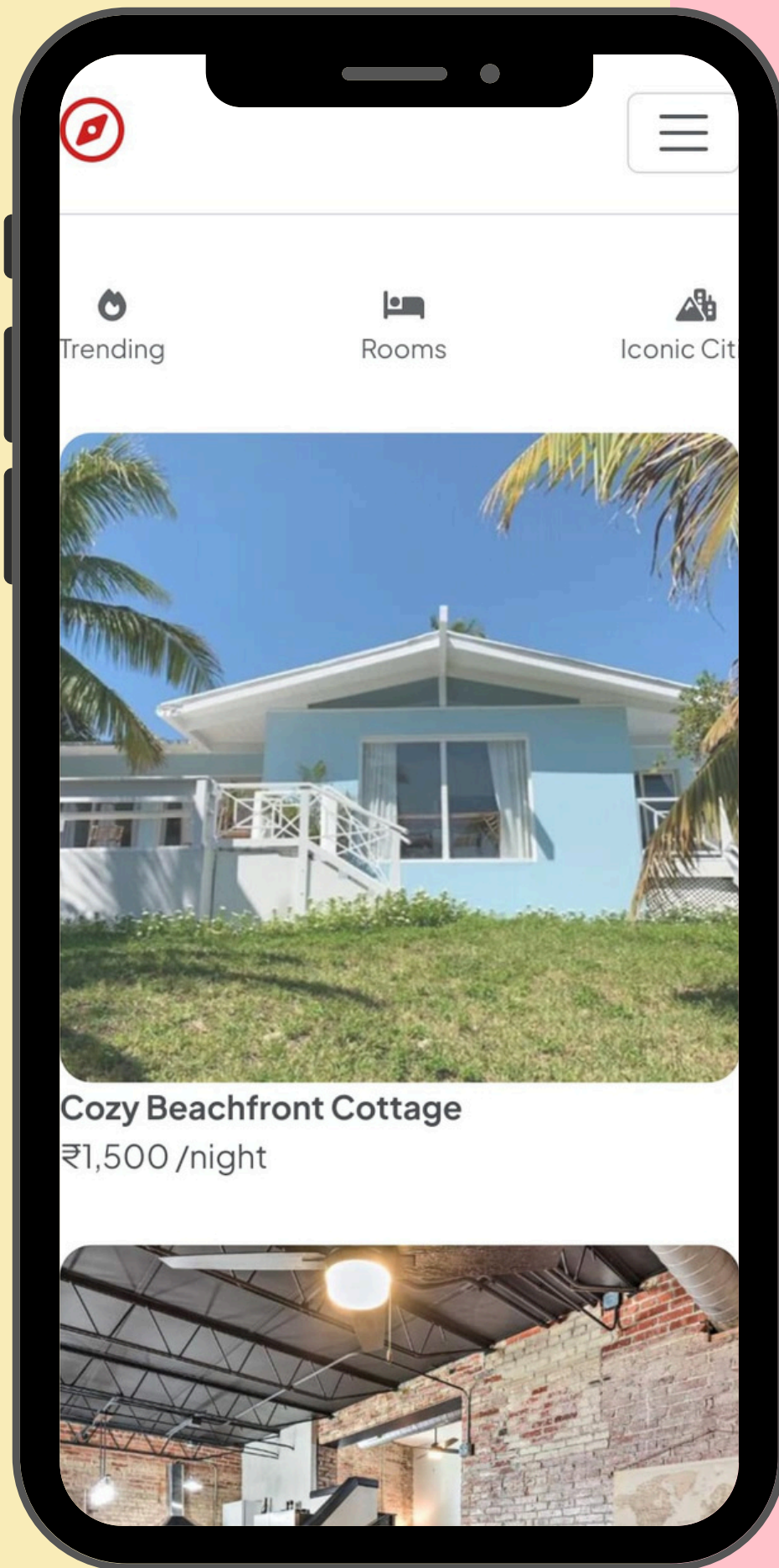
Cloudinary: A cloud-based service seamlessly integrated for storing and serving images, managing uploads, and optimizing media assets.

HTML/CSS: The backbone of the frontend, used to structure content and design the user interface, making the application visually appealing and easy to navigate.

EJS (Embedded JavaScript): A powerful templating language, allowing us to generate dynamic HTML with embedded JavaScript, enabling smooth data rendering on the frontend.

Render: Chosen for hosting the website, offering a reliable and scalable platform to deploy and manage the application in a production environment.





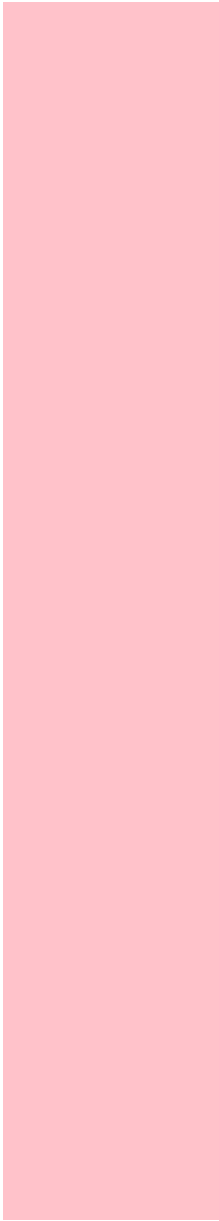
New System Benefits

Intuitive User Experience: Effortless search, tailored recommendations, and smooth booking.

Comprehensive Travel Planning: Explore destinations, create itineraries, and discover hidden gems with interactive maps and real-time data. Utilize map APIs for dynamic map displays, location-based search, and route optimization.

Instant Support: 24/7 chatbot assistance for queries and bookings.
Diverse Accommodation: A wide range of unique stays to suit every traveler.

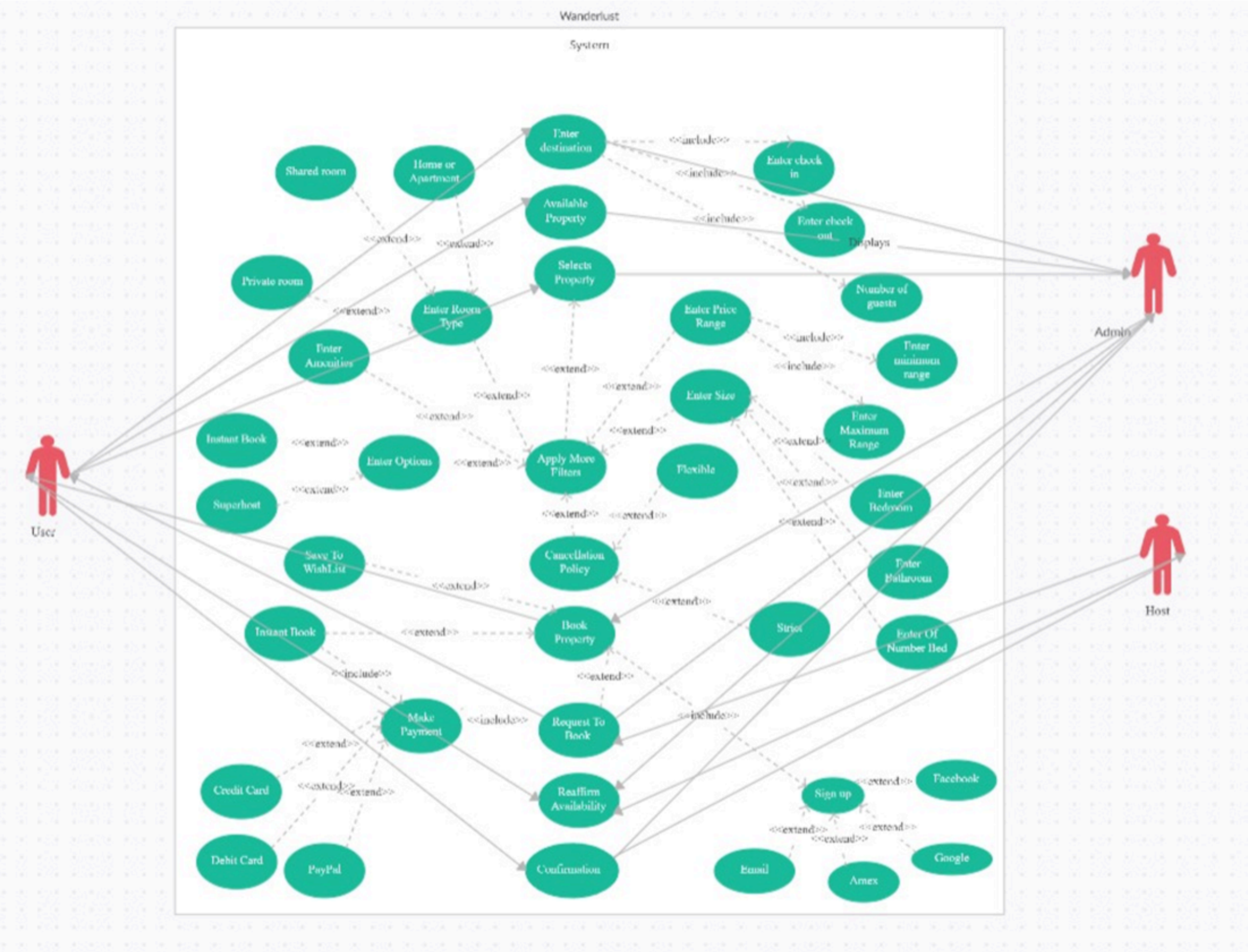
AI-Powered Personalization: Create custom itineraries based on preferences, budget, and interests. AI suggests optimal routes, accommodations, and activities, integrated with map APIs for visual guidance and interactive exploration.



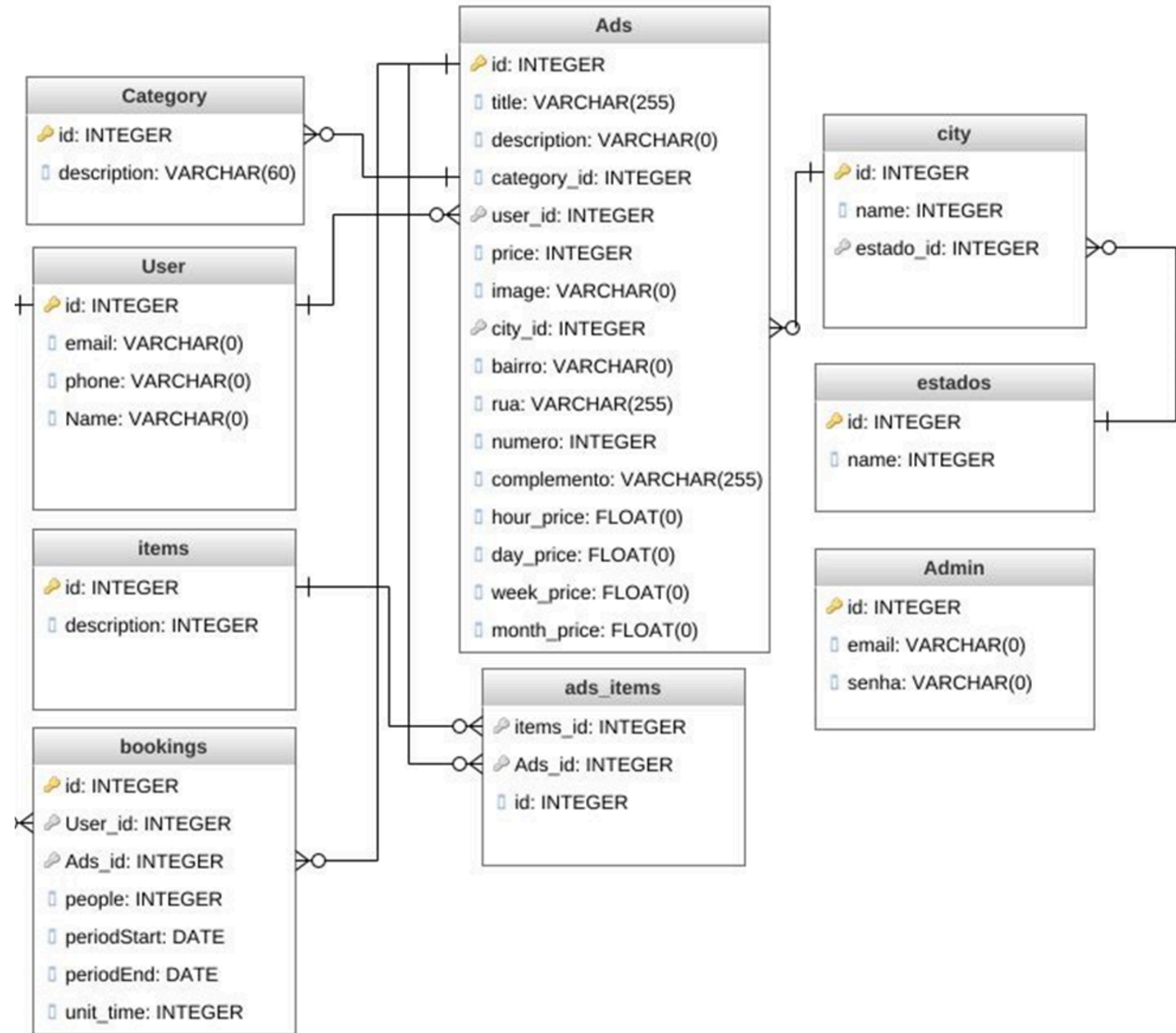
Thank You

UML Diagrams

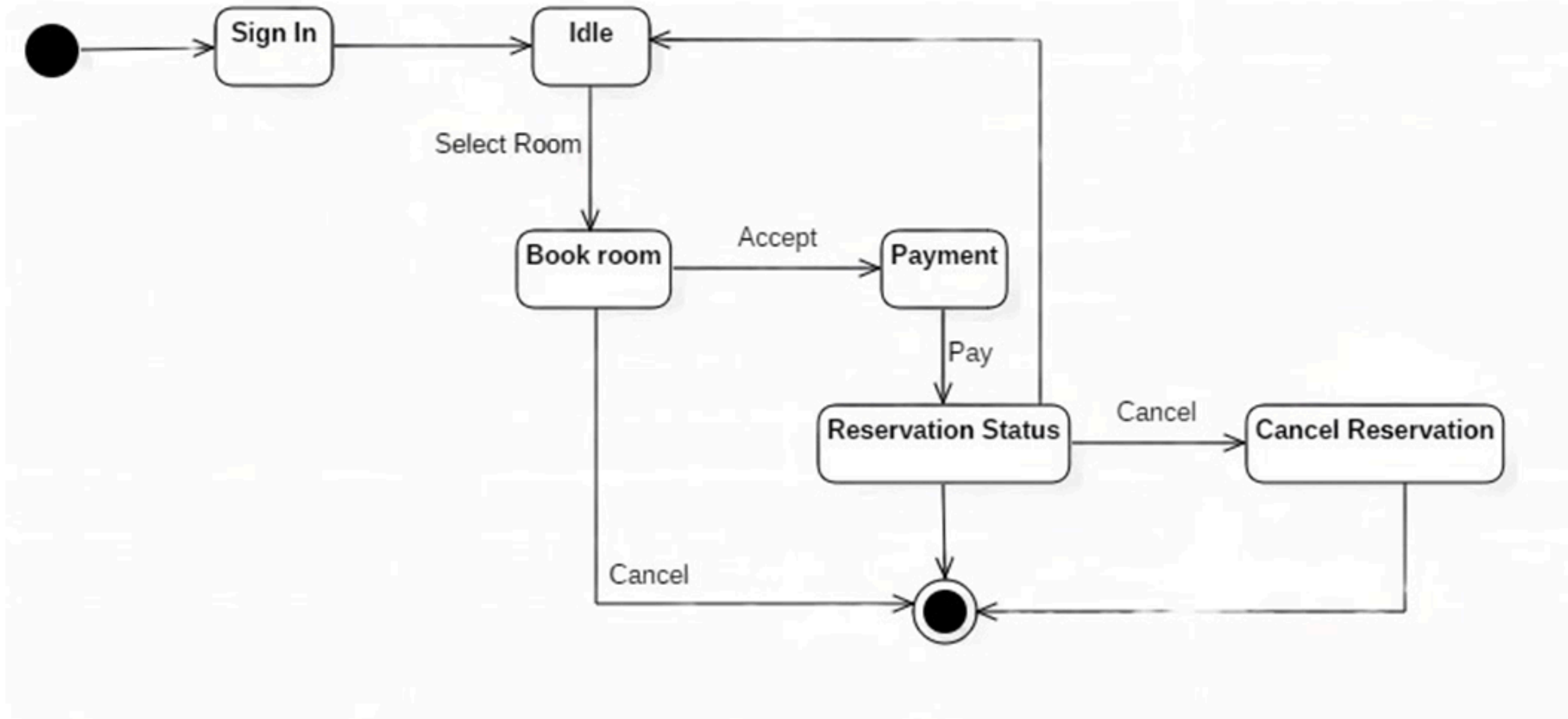
Use Case Diagram:



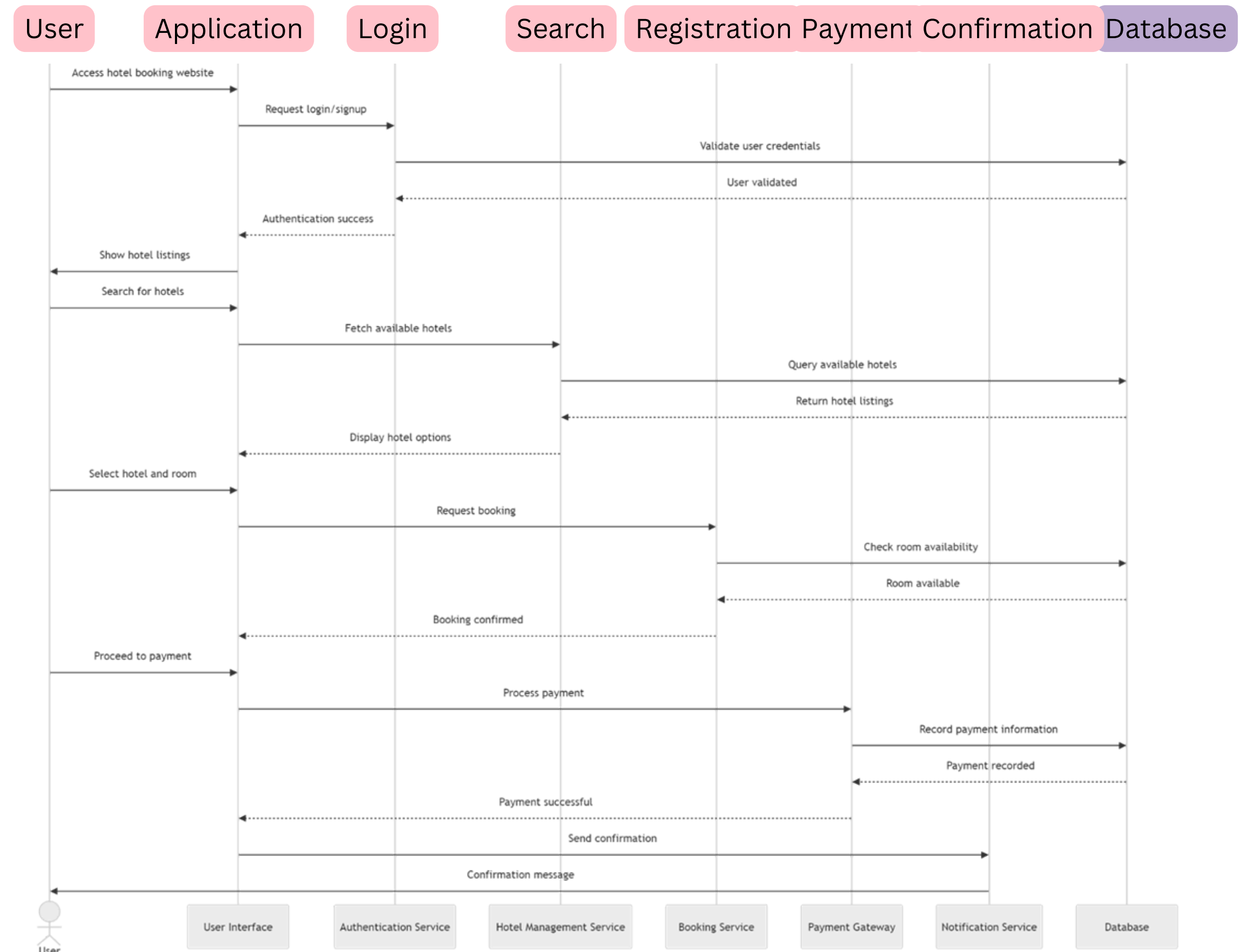
Class Diagram:



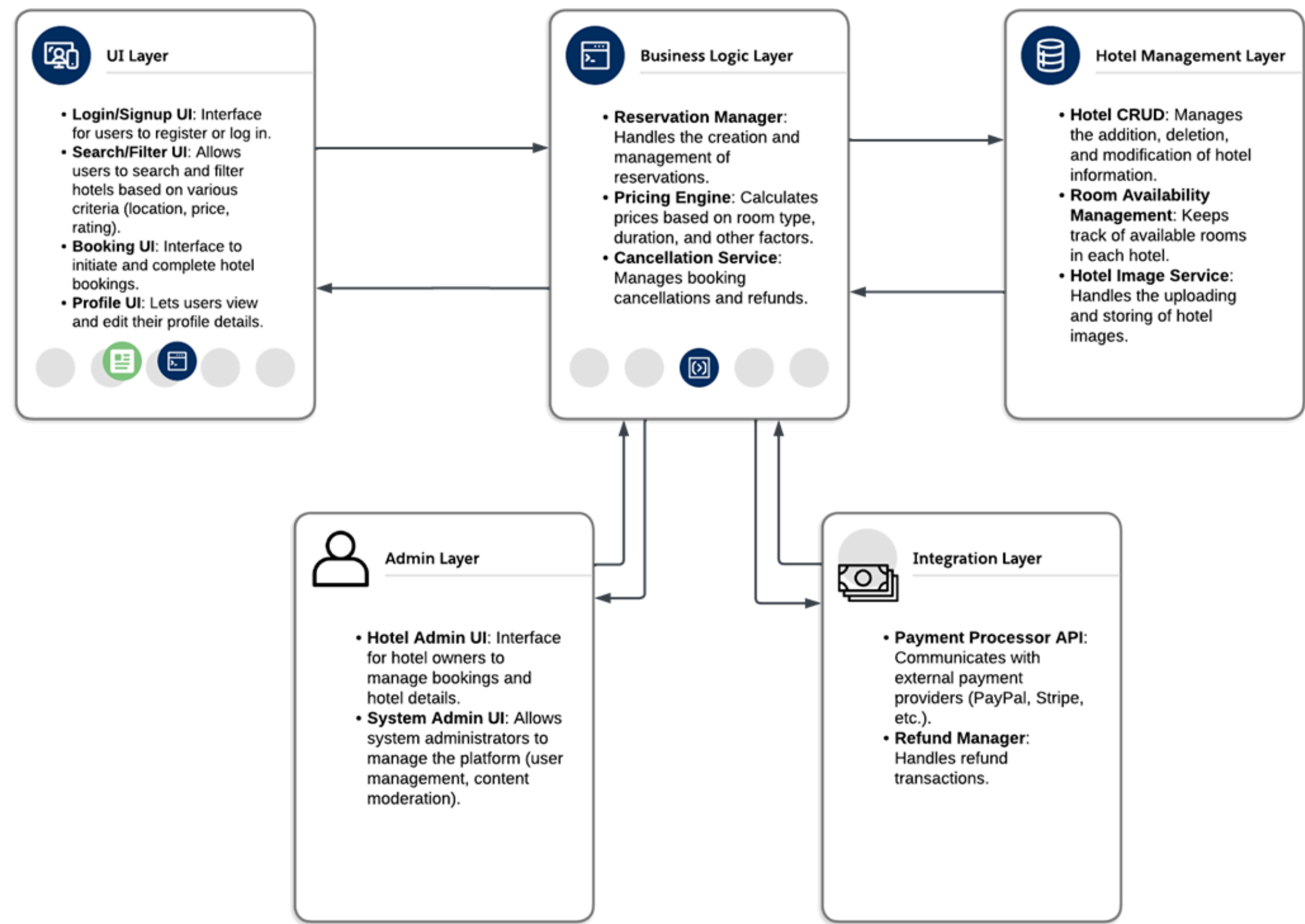
State Class Diagram:



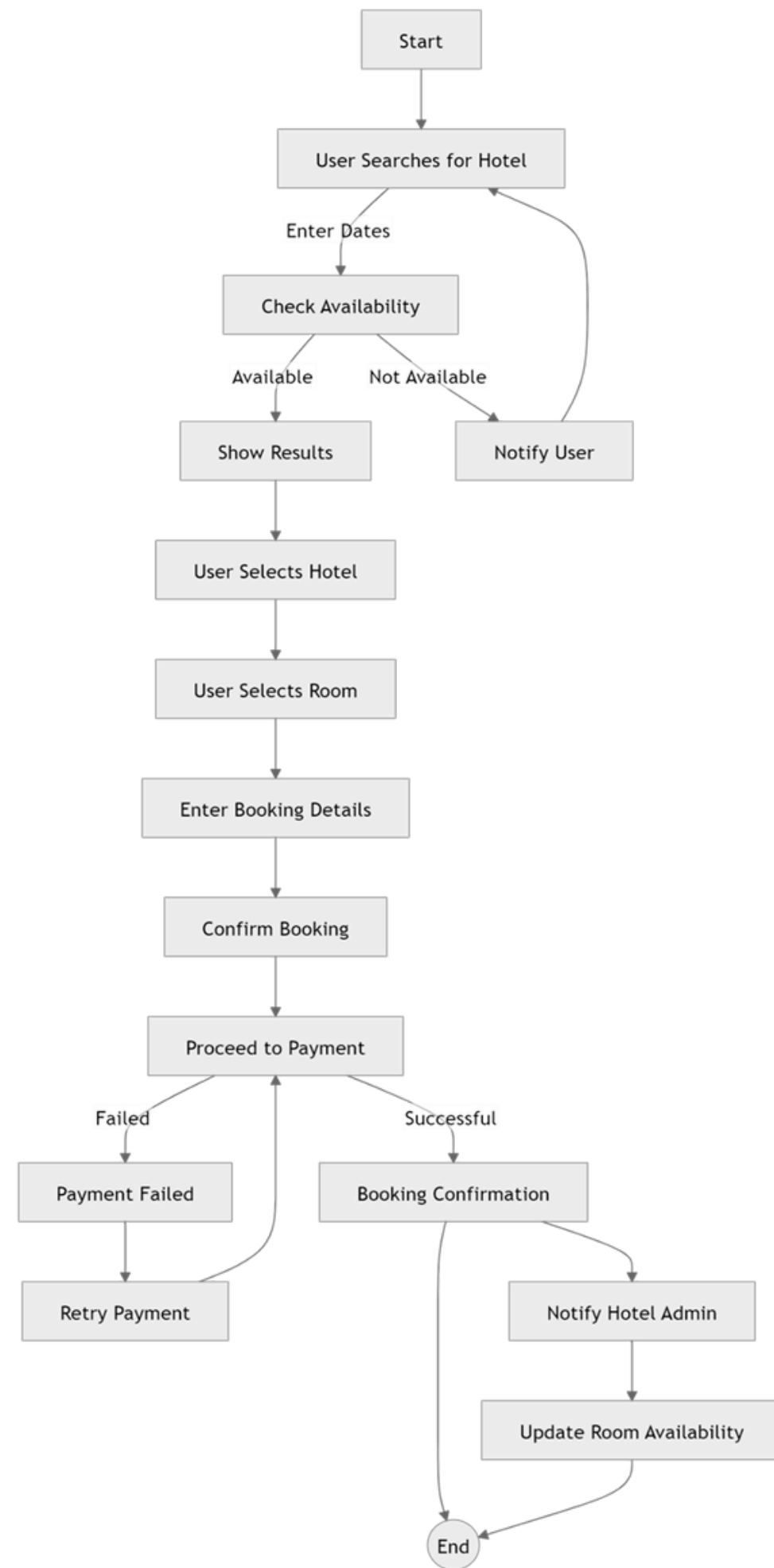
Sequence Diagram:



Component Diagram:

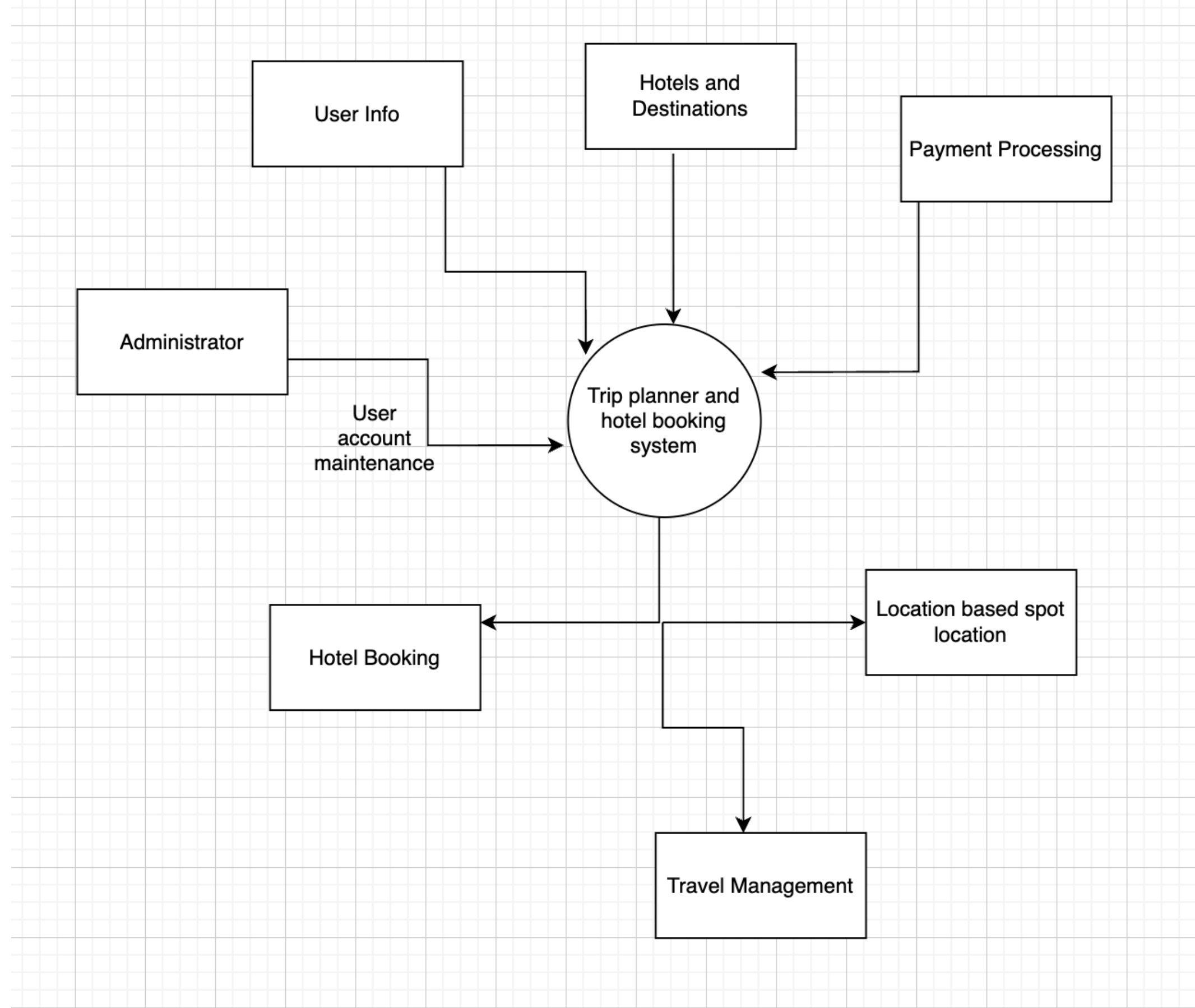


Activity Diagram:

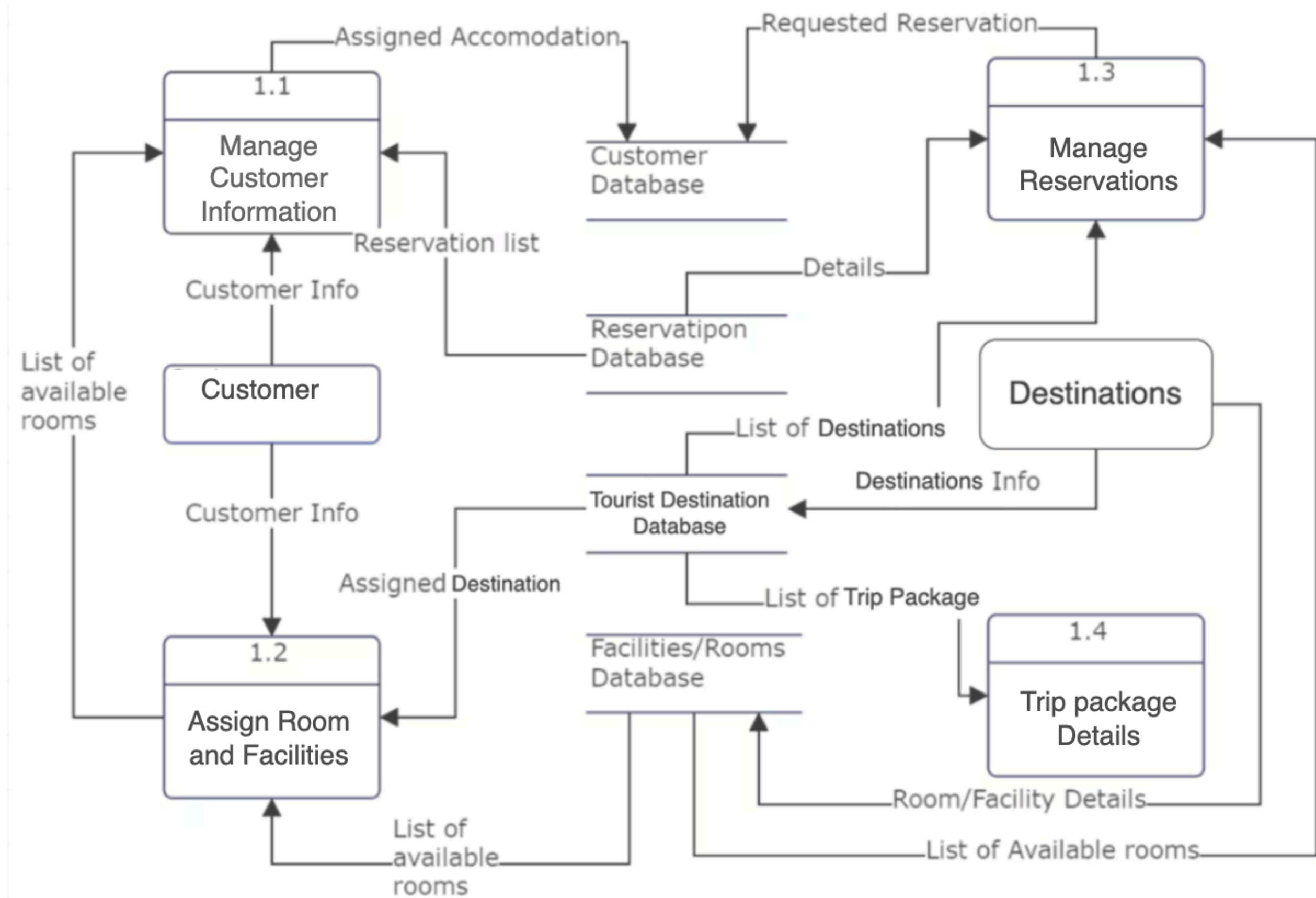


DFD Diagram:

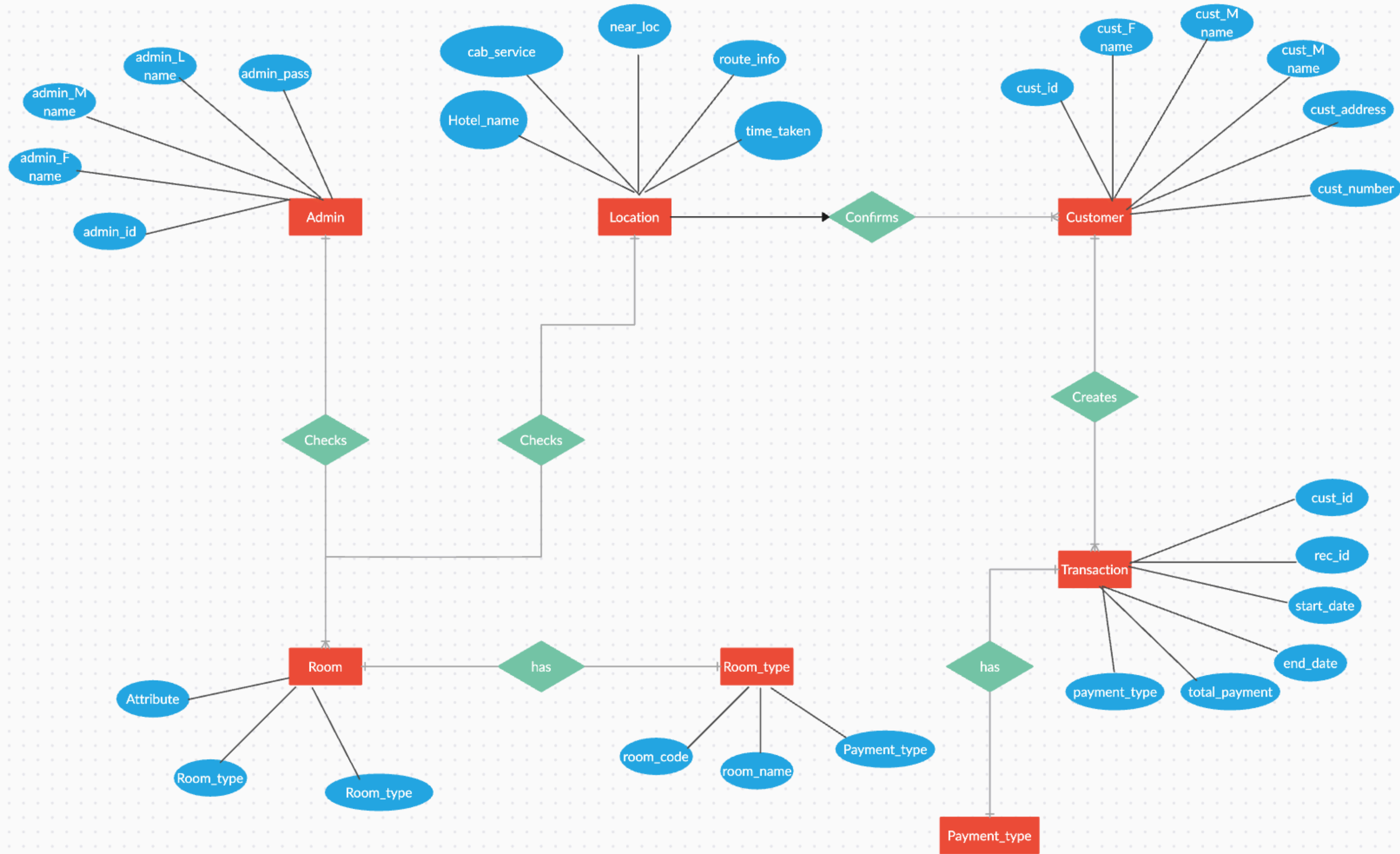
Level 0 DFD:



Level 1 DFD:



ER Diagram



Admin Table:

admin_id	F_name	M_name	L_name	admin_pass
1	John	A	Doe	password123
2	Alice	B	Smith	passAlice2024
3	Robert	NULL	Johnson	robertStrongPass
4	Mary	C	Brown	marySecurePass

Location:

location_id	Hotel_name	cab_service	near_loc	route_inf	time_taken
1	Grand Palace Hotel	1	Central Park	Via Main St and 5th Ave	30 minutes
2	Seaside Inn	0	Beachfront	Take the Coastal Highway	15 minutes
3	Mountain Lodge	1	Mount Hills	Drive through Forest Road	45 minutes
4	City Hotel	1	Downtown Plaza	Main Road and turn left at Plaza St	10 minutes

Customer:

cus_id	cus_Fname	cust_Mname	cust_Lname	cust_address	cust_number
1	John	A	Doe	123 Main St, New York, NY	1234567890
2	Alice	B	Smith	456 Elm St, Los Angeles, CA	9876543210
3	Robert	NULL	Johnson	789 Oak St, San Francisco, CA	4561237890
4	Mary	C	Brown	321 Pine St, Chicago, IL	3216549870

Transaction:

room_id	Room_type	Attribute
1	Deluxe	Sea view, Air Conditioning, King Bed, Balcony
2	Standard	City view, Double Bed, Free Wi-Fi
3	Suite	Mountain view, Private Pool, Jacuzzi, King Bed
4	Single	Single Bed, Free Wi-Fi, Garden view

Room_type:

room_code	room_name	payment_type
1	Deluxe Room	per_night
2	Standard Room	per_night
3	Suite Room	per_week
4	Single Room	per_month